Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

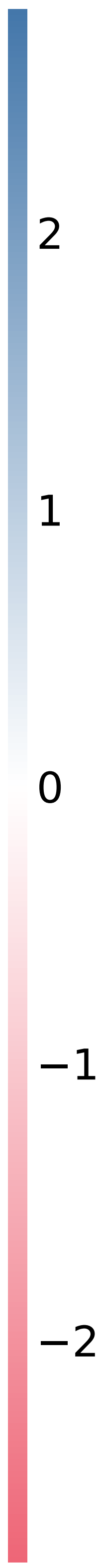
F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV



Reasoning + Answer:

<think>

Concise explanation:

- De Broglie wavelength of an electron with nonrelativistic kinetic energy T = p Â² / 2 m relates to momentum via Î» = h/p. From E = p Â² / 2 m, solve for p = sqrt(2 m E).

- With E = 1 eV = 1.602 ÃĹ 10 ^ âĪĴ 12 erg, and electron mass m = 9.109 ÃĹ 10 ^ âĪĴ 28 g, p = sqrt((2 ÃĹ 9.109 ÃĹ 10 ^ âĪĴ 28 g ÃĹ 1.602 ÃĹ 10 ^ âĪĴ 12 erg)/1 eV) âīĪ 1.31 ÃĹ 10 ^ âĪĴ 23 g Â· cm/s.

- Then Î» = h/p âīĪ (6.626 ÃĹ 10 ^ âĪĴ 27 erg Â· s) / (1.31 ÃĹ 10 ^ âĪĴ 23 g Â· cm/s) âīĪ 5.06 ÃĹ 10 ^ âĪĴ 4 cm, which is 5.06 ÃĹ 10 ^ âĪĴ 4 cm = 5.06 ÃĹ 10 ^ âĪĴ 3 mm âīĪ 0.0506 mm, about 0.51 Î¼m in micrometers.

Correct answer: J. 0.512 ÃĹ